Andrei Stefan Bejgu (he/him)

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Date of Birth: 16-11-1995
Location: Rome, Italy
Languages: English (Fluent), Italian (Native), Romanian (Native)

NLP professional with 7+ years of experience spanning software development, engineering, and research. I focus on multilingual NLP, retrieval-augmented generation (RAG), embedding-driven solutions and advanced Large Language Models (LLMs) fine-tuning – taking products from concept to large-scale deployment.

Experience & Project Selection

Babelscape *Rome, Italy*
NLP Researcher Oct. 2021–Present

I lead advanced NLP research that transforms state-of-the-art insights into innovative AI products, with a focus on retrieval-augmented systems powered by multilingual LLM fine-tuning. I work closely with multidisciplinary teams to integrate these solutions, while mentoring interns at every stage of the project lifecycle.

- Designed and developed real-time retrieval pipelines, maximizing precision and efficiency for Vera, a multilingual fact-checking system.
- **Distributed** (Slurm) fine-tuning of open-source LLMs for domain-specific tasks such as **claim extraction and verification**.
- Contributed to the safety fine-tuning of **Minerva**, the first Italian LLM.
- Engineered **Multilingual Embeddings** for RAG systems, driving state-of-the-art retrieval.
- Improved **Entity Linking** in Comprehendo, achieving a **12%** accuracy gain while reducing overhead by **60%**.

NLP Engineer Jun. 2020–Oct. 2021

I built robust, production-grade solutions by developing scalable pipelines that integrate technologies like Docker, Kubernetes, and Spark. I also implemented efficient data processing and indexing systems to ensure high performance and reliability across various AI products.

- Designed and deployed Semantic Paths, a **Multilingual Semantic Search** engine for scientific articles written in Kotlin.
- Developed advanced **Word Sense Disambiguation** models for large-scale multilingual processing.
- Optimized NLP systems for speed, accuracy, and scalability, improving overall robustness and reliability.

Software Engineer Jul. 2017–Jun. 2020

I engineered scalable, high-performance systems by designing and implementing multiprocess pipelines for enterprise-grade applications.

- Developed myKnowledgeGraph, a **Knowledge Graph Extraction** and visualization system for enterprise data.
- Designed the company's **Java NLP Pipeline**, maximizing speed, accuracy, and scalability of multilingual text processing.
- Enhanced **concept identification** in TraDeInterpret, increasing accuracy by **13%** in trademark analysis.

CINI - Consorzio Interuniversitario Nazionale per l'Informatica *Rome, Italy*
Software Developer Dec. 2016–Jan. 2017
Created an advanced tool for lexicon-semantic stylometric analysis, enabling efficient processing of large-scale texts.

Sapienza University of Rome *Rome, Italy*
Teaching Assistant Feb. 2022–Feb. 2023
Designed and led practical lab sessions and assignments for the Natural Language Processing Course, teaching master students state-of-the-art NLP techniques.

Education

Ph.D. in Artificial Intelligence

Sapienza University of Rome Jan. 2025

Focus: Explored multilingual NLP by applying lexical semantics to modern systems. Improved domain-specific tasks, looked into complex linguistic patterns, and worked on word- and sentence-level representations in RAG systems and LLMs. This was a joint industrial Ph.D., combining academic research with industry-driven projects. Created context-aware, domain-specific embeddings that boosted retrieval performance. –**Advisor:** Prof. R. Navigli

MSc in Computer Science

*Sapienza University of Rome –**Grade**: 110/110 with honors* Nov. 2021

Thesis: Exploring language-agnostic latent spaces for sentences. –**Advisor:** Prof. R. Navigli

BSc in Computer Science

*Sapienza University of Rome –**Grade**: 107/110* Jun. 2019

Thesis: Knowledge-Based Word Sense Disambiguation leveraging synset embeddings. –**Advisor:** Prof. R. Navigli

Publications

[1] A. Scirè*, **A. S. Bejgu***, S. Tedeschi, K. Ghonim, F. Martelli, and R. Navigli. "Truth or Mirage? Towards End-to-End Factuality Evaluation with LLM-OASIS". In: *Under Review*. 2025.

[2] L. Moroni, G. Puccetti, P.-L. H. Cabot, **A. S. Bejgu**, A. Miaschi, E. Barba, F. dell'Orletta, A. Esuli, and R. Navigli. "Adaptation of LLMs toward Italian Language: A Systematic Approach". In: *NAACL*. 2025.

[3] K. Ghonim, **A. S. Bejgu**, and R. Navigli. "Conceptpedia: Leveraging Traditional Approaches for Multimodal Image-Text Representation". In: *Under Review*. 2025.

[4] **A. S. Bejgu**, E. Barba, L. Procopio, A. Fernández-Castro, and R. Navigli. "Word Sense Linking: Disambiguating Outside the Sandbox". In: *ACL*. 2024.

[5] F. Molfese, **A. S. Bejgu**, S. Tedeschi, S. Conia, and R. Navigli. "CroCoAlign: A Cross-Lingual, Context-Aware and Fully-Neural Sentence Alignment System for Long Texts". In: *EACL*. 2024.

[6] F. Martelli, **A. S. Bejgu**, et al. "XL-WA: a Gold Evaluation Benchmark for Word Alignment in 14 Language Pairs". In: *CLiC-it*. 2023.

Scholarships and Awards

- Merit Scholarship Winner, LazioDiSCo.it 2018–2020
- Best Employee Award, Babelscape 2021